Parametric Equations: Differentiation

Questions 1–23

Question 1. By finding the cartesian equation of the curve, given by

$$x = t - 1 \qquad \text{and} \qquad y = 2t^2$$

find $\frac{dy}{dx}$ in terms of x.

Question 2. Find $\frac{dy}{dx}$ in terms of t for each of the following:

a) $x = t + 1$ and $y = -2t^2 + 2$
b) $x = t(t + 3)$ and $y = t^2(t + 1)$
c) $x = e^{2t}$ and $y = 2e^t$
d) $x = 4e^{2t}$ and $y = 3e^{-t} + 1$
e) $x = t - \cos t$ and $y = t - \sin t$

Question 3. Prerequisite knowledge for Q3: Quotient Rule Find $\frac{dy}{dx}$ in terms of t for $x = \frac{1}{1+t^2}$ and $y = t^2$.

Question 4. It is given that $x = t + 1$, $y = t^2$, where $t \geq 0$. Find the equation of the tangent to the curve when $t = 2$.

Question 5. It is given that $x = t^3$, $y = 2t$, where $t \geq 0$. Find the equation of the normal to the curve when $t = 3$.

Question 6. It is given that $x = e^t$, $y = \frac{1}{t}$, where $t > 0$. Find the equation of the normal to the curve when $t = 1$.

Question 7. For each of the following curves defined parametrically, find the coordinates of the stationary point(s) of the curve:

a) $x = t^4 + t$ and $y = 10t - 5t^2$ where $t \geq 0$
b) $x = 6t$ and $y = t^3 - 5t^2 + 3t$ where $t \geq 0$
c) $x = e^{2t}$ and $y = 6t^2 - \frac{4}{3}t^3 - 5t$ where $t \geq 0$

Question 8. A curve is defined by the parametric equations

$$x = 3 - t^3 \qquad \text{and} \qquad y = (t + 2)^2$$

for $-1 \leq t \leq 1$. Find

a) the coordinate of the point with the maximum y-value.
b) the coordinate of the point with the minimum y-value.
c) $\frac{dy}{dx}$ in terms of t.

Question 9. A curve is defined by the parametric equations

$$x = \frac{4}{t} \qquad \text{and} \qquad y = 3t - 1$$

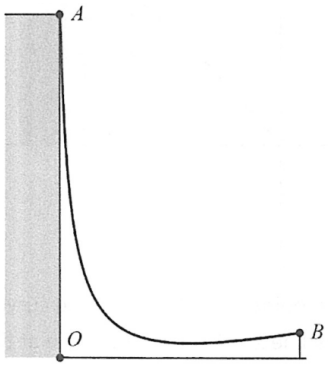
a) Find the equation of the tangent line to the curve at the point $(2, 5)$ giving your answer in the form $y = mx + c$.
b) Hence find the area bounded by the tangent line at $(2, 5)$ and the coordinate axes.

Question 10. Given that $t > 0$, find the equation of the normal to the curve given by

$$x = 1 - \frac{1}{t} \quad \text{and} \quad y = 2t(t - 1)$$

when $x = -2$, giving your answer in the form $ax + by = c$ where a, b and c are integers to be found.

Question 11. A drop slide is shown in the diagram. Point A is at the start of the slide and point is B at the end of the slide. Point O lies on the ground.



It may be modelled by parametric equations

$$x = \tfrac{1}{2}t^2 - \tfrac{1}{8} \quad \text{and} \quad y = \frac{5}{t} + t - \frac{21}{5}$$

for $0.5 \leq t \leq 3$. The horizontal distance from O is x metres and the vertical distance from O is y metres. According to the model, find

a) the height at the start of the slide.
b) the height at the end of the slide.
c) how far horizontally point B is from the start of the slide.
d) the difference in height between the start of the slide and the lowest point of the slide, giving your answer to the nearest centimetre.

Question 12. A curve is defined by the parametric equations

$$x = t + a \qquad \text{and} \qquad y = t^2 + 3$$

where $-3 \leq t \leq 3$ and a is an integer. Find

a) the cartesian curve which satisfies all the points on the curve.
b) the equation of the normal to the curve when $t = 1$ in the form $y = mx + c$, giving your answer in terms of a.

Question 13. A curve is defined by the parametric equations

$$x = \sqrt{3}\sin\theta \qquad \text{and} \qquad y = 2\sin 2\theta$$

for $0 \leq \theta \leq \frac{\pi}{2}$.

a) Find $\frac{dy}{dx}$ in terms of θ.

b) Show that the equation for the normal line to the curve at $\theta = \frac{\pi}{3}$ may be written in the form

$$a\sqrt{3}y - bx = c$$

where a, b and c are integers to be found.

c) Find the area bound by the coordinate axes and the normal line to the curve when $\theta = \frac{\pi}{3}$.

Question 14. A curve is defined by the parametric equations

$$x = 2 \times 3^{-t} + 4 \qquad \text{and} \qquad y = 4 \times 3^{t} - 6$$

for $t \in \mathbb{R}$.

a) Show that $\frac{dy}{dx} = -2 \times 3^{2t}$

b) Show that the curve does not cross the y-axis.

c) Show that the cartesian equation which satisfies all the points on the curve may be written in the form

$$y = \frac{a}{x - b} - c$$

where a, b and c are integers and $x > b$.

d) As $t \to \infty$, $x \to \alpha$. Find the value of α.

e) As $t \to -\infty$, $y \to \beta$. Find the value of β.

f) Find where the curve crosses the x-axis.

g) Hence sketch the curve.

Question 15. Prerequisite knowledge for Q15: Reciprocal Trigonometric Functions and Chain Rule A curve is defined by

$$x = \sin 4\theta \qquad \text{and} \qquad y = \sec^2 \theta$$

for $0 \leq \theta \leq \frac{\pi}{2}$.

a) Find $\frac{dy}{dx}$ in terms of θ.

b) Hence find the exact value of the gradient of the tangent of the curve when $y = 2$.

Question 16. A curve is defined by parametric equations

$$x = 5t^3 \quad \text{and} \quad y = t^4$$

for $t \geq 0$.

a) Show that the equation of the tangent at the point t on the curve is given by

$$15y - 4tx + 5t^4 = 0$$

A triangle is bound by the x-axis, the tangent line to the curve at $x = 40$ and the line $x = 40$.

b) Find the area of the triangle.

Question 17. A curve is defined by parametric equations

$$x = \tan\theta + 1 \qquad \text{and} \qquad y = \sec\theta$$

for $-\frac{\pi}{3} \leq \theta \leq \frac{\pi}{3}$. The point P lies on the curve when $\theta = a$ where $-\frac{\pi}{3} \leq a \leq \frac{\pi}{3}$.

a) Find the equation for the normal to the curve at the point P in the form $y = mx + c$ giving your answer in terms of a.

b) Hence find the minimum point on the curve.

Question 18. A curve is defined by

$$x = t^4 \quad \text{and} \quad y = 2t^3(1+t)$$

where $t \in \mathbb{R}$.

a) Find $\frac{dy}{dx}$ in terms of t.

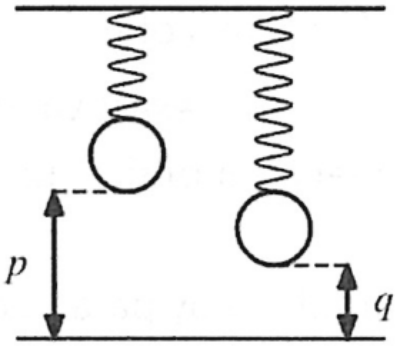
b) Find the coordinates of the minimum point of the curve.

c) Find where the curve crosses the y-axis.

d) Find where the curve crosses the x-axis.

e) Hence sketch the curve.

Question 19. Prerequisite knowledge for Q19–Q21: Harmonic forms Two springs are attached to a ceiling with masses attached. They oscillate up and down, between two heights above the floor, p and q metres as shown in the diagram.



They may be modelled using the parametric equations

$$p = \tfrac{1}{2} - \tfrac{1}{2}\cos t \quad \text{and} \quad q = \tfrac{1}{2} - \tfrac{1}{4}\sin t$$

where $0 \leq t \leq 8\pi$.

a) Find the initial height difference between the two springs.
b) Write the difference in height between them in the form $R\sin(t-\alpha)$ where R and α are positive constants to be found and $0 < \alpha < \frac{\pi}{2}$.
c) Find the maximum distance between the two springs.

Question 20. A curve is defined by

$$x = 4\cos\theta \quad \text{and} \quad y = 5\sin\theta$$

for $0 \leq \theta \leq 2\pi$. The tangent lines to the curve at two points, A and B, pass through the point $(4, 12)$.

a) Find $\frac{dy}{dx}$ in terms of θ.
b) State the equation of the tangent line in terms of θ.
c) Hence show that values of θ at A and B satisfy the equation

$$12\sin\theta + 5\cos\theta = 5$$

d) Hence find the two possible values of θ.

Question 21. A curve is defined by

$$x = a\cos\theta \quad \text{and} \quad y = 4a\sin\theta$$

a) Show that the equation of the tangent line is

$$y\sin\theta + 4x\cos\theta = 4a$$

The tangent line passes through the point $(a, 3a)$.

b) For $0 < \alpha < \frac{\pi}{2}$, show that θ satisfies

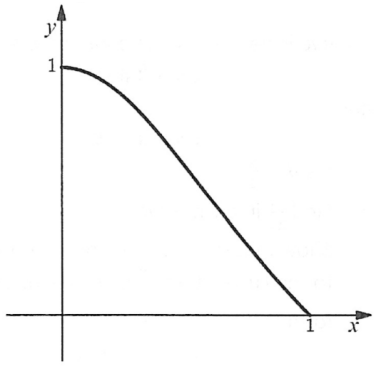
$$5\sin(\theta + \alpha) = 4$$

c) Hence find the two possible values of θ when $0 \leq \theta < 2\pi$.

Question 22. Prerequisite knowledge for Q22: The Newton-Raphson Method and Double Angle Formulae The curve C shown below is defined by parametric equations

$$x = \tan t \quad \text{and} \quad y = \cos 2t$$

for $0 \leq t \leq \frac{\pi}{4}$.



a) Find the equation of the line that connects the points on the curve C when $t = 0$ and $t = \frac{\pi}{4}$.

b) Show that $\frac{dy}{dx} = -4\sin t\cos^3 t$

c) Use Newton-Raphson's method starting at $x_0 = 0$ to show that a solution to

$$4\sin x\cos^3 x = 1$$

occurs when $x = 0.2874$ (to 4 significant figures).

d) Hence, find the range of values of k such that the line $y + x = k$ intersects the curve at least once.

Question 23. Prerequisite knowledge for Q23: Product Rule and Small Angle Approximations A curve is defined by the parametric equations

$$x = \tfrac{1}{8}\sin t \quad \text{and} \quad y = t\cos t$$

where $0 \leq t \leq \pi$.

a) Find $\frac{dy}{dx}$ in terms of t.

b) Given that the stationary point is close to the origin, find the stationary point of the curve by using small angle approximations for $\cos t$ and $\sin t$.